Bishwamittra Ghosh

Postdoctoral Researcher
MPI-SWS, Germany
✉ *bghosh@mpi-sws.org*
bishwamittra.github.io
scholar, dblp

Research Interests

My research applies formal methods to machine learning, focusing on the foundational understanding of large language models and building trustworthy machine learning systems through fairness and interpretability.

Appointments

Since 04.24 **Postdoctoral Researcher**, *Max Planck Institute for Software Systems*, Germany.
Advisor: Dr. Krishna P. Gummadi

06.23 – 03.24 **Scientist**, *Institute of High Performance Computing (IHPC), A*STAR*, Singapore.

01.18 – 06.23 **Graduate Research Assistant**, *National University of Singapore (NUS)*, Singapore.

10.17 – 01.18 **Lecturer**, *Department of CSE, United International University (UIU)*, Bangladesh.

Education

01.18 – 09.23 **PhD in Computer Science**, *National University of Singapore (NUS)*, Singapore.
Thesis: Interpretability and Fairness in Machine Learning: A Formal Methods Approach
Advisor: Dr. Kuldeep S. Meel

02.13 – 09.17 **BSc in Computer Science and Engineering**, *Bangladesh University of Engineering and Technology (BUET)*, Bangladesh, CGPA: 3.83 out of 4 (with Honours).
Thesis: The Flexible Socio Spatial Group Queries
Advisor: Dr. Mohammed Eunus Ali

Publications (reverse chronological order)

Conference Papers

NeurIPS-26 Understanding the Interplay between Memorization and Learning in Large Language Models
Bishwamittra Ghosh, Soumi Das, Qinyuan Wu, Mohammad Aflah Khan, Deepak Garg, Krishna P. Gummadi, Evimaria Terzi

ACL-26 Fine-tuning vs. In-context Learning in Large Language Models: A Formal Language Learning Perspective (SAC highlight)
Bishwamittra Ghosh, Soumi Das, Till Speicher, Qinyuan Wu, Mohammad Aflah Khan, Deepak Garg, Krishna P. Gummadi, Evimaria Terzi

ICLR-26 Rote Learning Considered Useful: Generalizing over Memorized Data in LLMs
Qinyuan Wu, Soumi Das, Mahsa Amani, Bishwamittra Ghosh, Mohammad Aflah Khan, Krishna P. Gummadi, Muhammad Bilal Zafar

ICLR-26 In Agents We Trust, but Who Do Agents Trust? Latent Preferences Steer LLM Generations
Mohammad Aflah Khan, Mahsa Amani, Soumi Das, Bishwamittra Ghosh, Qinyuan Wu, Krishna P. Gummadi, Manish Gupta, Abhilasha Ravichander

ICLR-25 Logical Consistency of Large Language Models in Fact-checking
Bishwamittra Ghosh, Sarah Hasan, Naheed Anjum Arafat, Arijit Khan

AAAI-25 Active Fourier Auditor for Estimating Distributional Properties of ML Models
Ayoub Ajarra, Bishwamittra Ghosh, Debabrota Basu

ECAI-25 History-Aware and Dynamic Client Contribution in Federated Learning
Bishwamittra Ghosh, Debabrota Basu, Huazhu Fu, Yuan Wang, Renuga Kanagavelu, Jin Peng Jiang, Yong Liu, Rick Siow Mong Goh, Qingsong Wei

WSDM-25 Towards Reliable Latent Knowledge Estimation in LLMs: Zero-Prompt Many-Shot Based Factual Knowledge Extraction
Qinyuan Wu, Mohammad Aflah Khan, Soumi Das, Vedant Nanda, Bishwamittra Ghosh, Camila Kolling, Till Speicher, Laurent Bindschaedler, Krishna P. Gummadi, Evimaria Terzi

CAI-24 Split Learning of Multi-Modal Medical Image Classification
Bishwamittra Ghosh, Yuan Wang, Huazhu Fu, Qingsong Wei, Yong Liu, Rick Goh

FAccT-23 "How Biased are Your Features?": Computing Fairness Influence Functions with Global Sensitivity Analysis
Bishwamittra Ghosh, Debabrota Basu, Kuldeep S. Meel

VLDB-23 Neighborhood-based Hypergraph Core Decomposition
Naheed Anjum Arafat, Arijit Khan, Arpit Kumar Rai, Bishwamittra Ghosh

AAAI-22 Algorithmic Fairness Verification with Graphical Models (Oral)
Bishwamittra Ghosh, Debabrota Basu, Kuldeep S. Meel

AAAI-21 Justicia: A Stochastic SAT Approach to Formally Verify Fairness
Bishwamittra Ghosh, Debabrota Basu, Kuldeep S. Meel

AAAI-20 A MaxSAT-based Framework for Group Testing (Oral)
Lorenzo Ciampiconi, Bishwamittra Ghosh, Jonathan Scarlett, Kuldeep S. Meel

ECAI-20 Classification Rules in Relaxed Logical Form
Bishwamittra Ghosh, Dmitry Malioutov, Kuldeep S. Meel

AIES-19 IMLI: An Incremental Framework for MaxSAT-Based Learning of Interpretable Classification Rules
Bishwamittra Ghosh, Kuldeep S. Meel

VLDB-18 The Flexible Socio Spatial Group Queries
Bishwamittra Ghosh, Mohammed Eunus Ali, Farhana M. Choudhury, Sajid Hasan Apon, Timos Sellis, Jianxin Li

Journal Papers

JAIR-22 Efficient Learning of Interpretable Classification Rules
Bishwamittra Ghosh, Dmitry Malioutov, Kuldeep S. Meel

TSAS-21 Social-Spatial Group Queries with Keywords
Sajid Hasan Apon, Mohammed Eunus Ali, Bishwamittra Ghosh, Timos Sellis

Professional Experiences

Tutorials

IJCAI-23 Auditing Bias of Machine Learning Algorithms: Tools and Overview

Talks

Invited Talks Towards Foundational Understanding and Trustworthiness of Machine Learning: A Formal Methods Approach
- Texas State University (26), SUTD (26), Dagstuhl Seminar (26, 25), Saarland University (25), MPI-SP (25), IIT-Delhi AD (25)

Trustworthy Machine Learning: A Formal Methods Approach
- Microsoft Research (24), University of Michigan, Dearborn (23), MPI-SWS, Germany (22), INRIA, France (22), IHPC, Singapore (22)

Conference Talks ACL (26), ICLR (25), FAccT (23), AAAI (22, 21, 20), ECAI (20), CP (19), VLDB (19), IJCAI (19), AIES (19)

Program Committee

AAAI (26, 25)

Reviewer

NeurIPS, ICML, ICLR, AAAI, TMLR, KDD, JAIR, AIES, CAV, TNNLS, TMI, TKDE

Internships

03.23 – 05.23 Max Planck Institute For Software Systems, Saarbrücken, Germany

11.21 – 02.22 INRIA, Lille-Nord, France

06.21 – 08.21 Goldman Sachs, Bangalore, India

03.20 – 06.20 Max Planck Institute For Software Systems, Kaiserslautern, Germany

Teaching Service

10.26 – 04.27 Abilities and Limitations of LLMs: A Formal Language Learning Perspective, *RPTU*, Germany
06.18 – 06.21 CS2030: Programming Methodology II, *NUS*, Singapore
06.20 – 12.20 CS3243: Introduction to Artificial Intelligence, *NUS*, Singpapore
Fall'17 Numerical Methods, *UIU*, Bangladesh
Fall'17 Data Communication, *UIU*, Bangladesh

Awards and Honors

2026 SAC Highlight, ACL 2026
2026 Invited for Dagstuhl Seminar on User-Aligned Assessment of AI Systems
2025 Invited for Dagstuhl Seminar on Interactions in Constraint Optimization
2025 KDD 2025 Outstanding reviewer (top 10% of reviewers)
2021 Moblilex Scholarship, Université de Lille, France
2018 – 2023 NUS Research Scholarship, Singapore
2013 – 2017 University Dean's List, BUET, Bangladesh
2011, 2012 National Math Olympiad winner, Bangladesh
2012 National Chemistry Olympiad winner, Bangladesh

References

- **Krishna P. Gummadi**
  Scientific Director
  Max Planck Institute for Software Systems (MPI-SWS), Germany

- **Kuldeep S. Meel**
  Associate Professor
  School of Computer Science
  Georgia Institute of Technology, USA

- **Evimaria Terzi**
  Professor and Associate Chair for Academics
  Computer Science Department MCS 280
  Boston University, USA

- **Debabrota Basu**
  Faculty (equivalent of Associate Professor)
  Scool Team
  INRIA at Université de Lille and CNRS, France

- **Arijit Khan**
  Associate Professor
  Department of Computer Science
  Bowling Green State University, USA

Last update: October 13, 2026
Latest version: `https://bishwamittra.github.io/resume/bghosh_CV.pdf`